

3.8 The Root Test

At times the ratio test will be difficult to use since it requires the calculation of a limit that may be difficult to determine. In such situations this next test can be useful.

Theorem (The Root Test): Consider the series $\sum_{k=1}^{\infty} a_k$ and let $\rho = \lim_{k \rightarrow \infty} \sqrt[k]{|a_k|} = \lim_{k \rightarrow \infty} (|a_k|)^{1/k}$.

1. If $\rho < 1$, the series converges absolutely.
2. If $\rho > 1$ or $\rho = +\infty$, the series diverges.
3. If $\rho = 1$, the test is inconclusive and no conclusion can be drawn.

We will omit the proof of this theorem.

Example 1: Determine (with justification) whether the series $\sum_{k=1}^{\infty} \frac{1}{(\ln(k+1))^k}$ converges or diverges.

We note that $\sum_{k=1}^{\infty} \frac{1}{(\ln(k+1))^k}$ contains only positive terms and so we calculate:

$\rho = \lim_{k \rightarrow \infty} \left(\frac{1}{(\ln(k+1))^k} \right)^{1/k} = \lim_{k \rightarrow \infty} \left(\frac{1}{\ln(k+1)} \right) = 0 < 1$. Therefore, by the root test, the series $\sum_{k=1}^{\infty} \frac{1}{(\ln(k+1))^k}$ converges. ■

Example 2: Determine (with justification) whether the series $\sum_{n=1}^{\infty} \left(\frac{-2n}{n+1} \right)^{5n}$ converges or diverges.

We calculate that $\lim_{n \rightarrow \infty} \sqrt[n]{\left| \left(\frac{-2n}{n+1} \right)^{5n} \right|} = \lim_{n \rightarrow \infty} \sqrt[n]{\left| \left(\frac{-2n}{n+1} \right)^5 \right|^n} = \lim_{n \rightarrow \infty} \sqrt[n]{\left| \frac{-32n^5}{(n+1)^5} \right|^n} = \lim_{n \rightarrow \infty} \sqrt[n]{\frac{|-32n^5|^n}{((n+1)^5)^n}}$

$$= \lim_{n \rightarrow \infty} \sqrt[n]{\left(\frac{32n^5}{(n+1)^5} \right)^n} = \lim_{n \rightarrow \infty} \frac{32n^5}{(n+1)^5} = \lim_{n \rightarrow \infty} \left(\frac{2n}{n+1} \cdot \frac{1/n}{1/n} \right)^5 = \left(\lim_{n \rightarrow \infty} \frac{2}{1+1/n} \right)^5 = 2^5 = 32 > 1.$$

Thus, the series diverges by the root test. ■

Example 3: Determine (with justification) whether the series $\sum_{k=1}^{\infty} \left(\frac{4k-5}{2k+1} \right)^k$ converges or diverges.

We note that $\sum_{k=1}^{\infty} \left(\frac{4k-5}{2k+1} \right)^k = -\frac{1}{3} + \sum_{k=2}^{\infty} \left(\frac{4k-5}{2k+1} \right)^k$. We can then see that the series $\sum_{k=2}^{\infty} \left(\frac{4k-5}{2k+1} \right)^k$ contains only

positive terms, so we calculate: $\rho = \lim_{k \rightarrow \infty} \left(\left(\left| \frac{4k-5}{2k+1} \right| \right)^k \right)^{1/k} = \lim_{k \rightarrow \infty} \left(\frac{4k-5}{2k+1} \right) = 2 > 1$.

So, $\sum_{k=2}^{\infty} \left(\frac{4k-5}{2k+1} \right)^k$ diverges and therefore the series $\sum_{k=1}^{\infty} \left(\frac{4k-5}{2k+1} \right)^k$ diverges.

■